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The Multiple-Access Channel with Partial State Information at the Encoders

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Abstract

This work studies the multiple access channel controlled by random parameters, with full side information at the decoder, and partial, rate limited, side information (SI) at the encoders. A single letter characterization of the capacity region is derived, for the special case where the side information is degraded. Here *degraded* side information refers to the case where the SI available at one of the encoders is a subset of the SI available at the other decoder. Inner and outer bounds are derived on the capacity region of that channel, for the general case where there are no restrictions on the structure of the SI at the two encoders. The techniques employed for coding the rate-limited SI, and the achievable regions so obtained, are closely related to the problems of hierarchical source coding, and multiple descriptions.

Index terms — Hierarchical coding, multiple access channel, multiple description, side information, state controlled channels, successive refinement.

1 Introduction

Channels that are controlled by random states are widely studied from both information theoretic and communications aspects. Beside their theoretical importance, channels with random parameters model faithfully the situation in practical systems, where the random

states represent random fading, interference imposed by adjacent users, etc. For a comprehensive overview of the field of state controlled channels, in the context of communications in random fading environments, see [3]. In many cases, side information about the channel state (CSI) is available at the transmitter (CSIT), and/or at the receiver (CSIR). The way CSIT is utilized depends, among other factors, on whether it is available to the encoder in a causal or non-causal manner. Since decoding at the channel output is done only after the whole codeword is received, distinction between the causal and non-causal case is relevant only at the encoder. Encoding for a single user state controlled channel with causal CSIT was first studied by Shannon [25]. Specifically, let $P_{Y|X,S}$ stand for the channel transition probability, where Y is the channel output, X the channel input, and S the random state, with distribution P_S, known causally at the encoder. Further, let $t: \mathcal{S} \to \mathcal{X}$ be a *strategy*, i.e., a mapping from the state space $\mathcal{S}$ to the input alphabet $\mathcal{X}$. Shannon showed that the capacity of this channel is given by

$$C = \max_{q(t)} I(T;Y), \tag{1}$$

where the maximization is over all distributions $q(\cdot)$ on the space of strategies. In [24], Salehi directly derived the capacity of single user channel for the case of *partial* CSIT and CSIR, i.e., the case where the encoder and decoder have access to noisy versions of the state random variables. Note that CSIR can be considered as part of the channel output Y, and thus, as far as capacity formulas are concerned, need not be treated separately. Caire and Shamai showed in [4] that the results of Salehi can be obtained from Shannon's result (1). Moreover, it was shown that for the special case that the CSIT is a subset of the CSIR, capacity can be achieved by coding over the input alphabet of the channel, without recourse to random coding over the space of strategies.

The study of coding for channels with non-causal CSIT was started with the works of Kusnetsov and Tsybakov [18], who considered such models in connection with coding for computer memories with defective cells, where the location of the defective cells is known a priori to the encoder. Their work started a series of papers studying coding schemes for computer memories (see, e.g., [27], [17] and refrences therein). The capacity of a general discrete memoryless channel (i.e., not necessarily a computer memory model) with non-causal CSIT was characterized by Gel'fand and Pinsker in [13]. Costa [5] applied the formula obtained by Gel'fand and Pinsker to the special model of Gaussian channel with additive

Gaussian interference, and showed that once the interference is known non-causally to the encoder, no loss in capacity is incurred relative to the case of no interference. Although it is initially stated for CSIT only, other models where the CSIT is a random map of the channel state, and the decoder is also equipped with side information, possibly distinct from the CSIT, can be obtained as simple special cases of the Gel'fand and Pinsker formula, as shown in [6].

In many practical scenarios, the encoder and decoder rarely have the ability to measure directly, or estimate, the channel state. Instead, a third party (e.g. a central station in cellular communication) has access to channel information, and conveys the CSI to its destinations (encoder and/or decoder) by means of wayside channels, or extra links dedicated to system information. Similarly, in applications involving computer memories with defects, information about the defective cells must be stored in memory for the potential use by the encoder or decoder. This means that system resources — bandwidth, time slots, storage space, etc. — have to be allocated to CSI, if one wishes to utilize it in the coding/decoding scheme. In such situations it is imperative to examine the utilization of CSI under rate constraints. The fact that even partial knowledge of the channel state can significantly increase performance, has been observed in connection with practical systems – see, e.g., [28], [2], [21] and references therein. These works suggest various forms of feedback, where the type of information to transfer to the transmitter is determined ad hoc. In [14], Heegard and El Gamal suggested a communication model where coded CSI is provided to the encoder and decoder of a single user channel, by means of rate limited, noiseless channels. Specifically, let us denote the rate limits of CSIT and CSIR by R_e and R_d, respectively, and by R the communication rate over the main (single user) channel (we will sometimes use the term *forward rate* for R). In [14, Theorem 1], Heegard and El Gamal derived an inner bound on the region of achievable triplets (R, R_e, R_d), where the coded CSIT, of rate R_e, is coded and provided to the encoder in a non-causal manner. From this inner bound, a lower bound on the capacity of the channel under rate constraints R_e and R_d is readily obtained. Their lower bound turns to be tight for the following special cases: (a) no state information to either encoder and decoder ($R_e = R_d = 0$), (b) complete state information at both sides ($R_e \geq H(S)$, $R_d \geq H(S|Y)$), (c) complete state information at the encoder and no state information at the decoder ($R_d = 0$ and $R_e = H(s)$), which corresponds to the Gel'fand and Pinsker [13] result, (d) rate limited state information at the encoder and complete state information at the decoder (R_e arbitrary, $R_d \geq H(S|Y)$). In [14, Theorem 2], there is an error in the evaluation

of case (d) from Theorem 1 there. It was later corrected by Rosenzweig *et. al.* in [22] and [23], who showed that when the decoder is fully informed about the state of the channel, and the CSIT is rate limited to R_e, the capacity of the forward channel is given by

$$C(R_e) = \max I(X;Y|S,S_0) \tag{2}$$

where the maximization is over all joint probability distributions P_{SXYS_0} of the form

$$P_{SXYS_0}(s,x,y,s_0) = P_S(s)P_{S_0|S}(s_0|s)P_{X|S_0}(x|s_0)P_{Y|XS}(y|x,s) \tag{3}$$

such that

$$R_e \geq I(S;S_0), \tag{4}$$

and S_0 is an auxiliary random variable whose alphabet size is bounded as

$$|\mathcal{S}_0| \leq |\mathcal{S}| + 1. \tag{5}$$

While many results about capacity of single-user channels with CSIT and CSIR appear in the literature, much less is known about their network counterparts. One exception is the multiple access channel (MAC) with causal CSIT. In [9], Das and Narayan characterized the capacity region of that channel, with various degrees of causal CSIT at the encoders. Their model accomodates also cases of noisy or delayed CSI, and the CSI available to the encoders and decoder may all be different. In addition, results are provided for the Gaussian fading MAC with causal CSIT and perfect CSIR on the fading, which is, in general, a stationary and ergodic process.

The random state controlled MAC with non-causal CSIT (a MAC extension of the Gel'fand and Pinsker channel) has not been solved yet. Its capacity region is known only for limited cases - e.g., the Gaussian MAC with additive Gaussian interference. Specifically, in [15], Kim, Sutivong, and Sigurjónsson derive the capacity region of the MAC extension of Costa's model, that is, the MAC with additive Gaussian interference known perfectly at the encoders. As in Costa's model, it turns out that once the interference is known non-causally at the encoders, no loss in capacity is incurred relative to the case of no intereference. This result can be proved either by substituting a specific input distribution in known achievabil-

ity formula, or by employing a coding technique based on Costa's ideas for the single-user channel. Either way, once an achievabilty result is obtained that coincides with the capacity region without interference, no converse has to be proved. In addition to the Gaussian MAC, Kim, Sutivong, and Sigurjónsson solve in [15] the Gaussian broadcast channel and the Gaussian physically degraded relay channel, with additive Gaussian interference and non-causal CSIT.

In this work we study the MAC extension of the model treated in [22]. We consider two basic models. The first, depicted in Figure 1, is the MAC with degraded partial CSIT and full CSIR. Specifically, a MAC is controlled by a random state sequence S^n, which is fully known at the decoder. We wish to transmit messages, at rates R_1 and R_2, across the MAC. The message indices in Figure 1 are denoted by w_1 and w_2, respectively. A third party, sometimes refered to as the *genie*, has access to the state sequece. The genie produces two streams of coded information about the state. The first stream, of rate R_e, is intended to the two users at the channel inputs, and hence its name, *common partial CSIT*. The term *partial* refers to the fact that the CSIT is a coded (rate-limited) version of the state sequence S^n and hence does not provide a full description of it. The second stream, of rate R_Δ, is intended to only one of the users, say user 2. Since the CSI available at encoder 1 (the encoder of the first user) is a subset of the CSI availabe at encoder 2, we refer to this model as MAC with *degraded partial CSIT* and full CSIR. The assumption that the CSIT has a degraded structure is motivated by practical situations, where there is a physical channel, with one common input, between the genie and the channel encoders, Encoder 1 and Encoder 2. In such a case, the genie faces a problem of transmitting a source S to the two users at the MAC inputs, via a broadcast channel. In many cases it is natural to further assume that the physical (broadcast) channel between the genie and the users is a degraded (e.g., when it is Gaussian), and therefore, one of the users gets a subset of the information provided to the second, stronger user. We will not treat here the complete model, where there is a noisy broadcast channel between the genie and the MAC inputs, as this involves joint source-channel coding for the degraded broadcast channel. Instead, we will adhere to the degraded structure of the CSIT, which closely describes the situation where the genie employs a separation strategy: i.e., it first encodes the source into two streams, of degraded staructure, and then uses a channel code designed for the degraded broadcast channel. Note that joint source-channel coding for the degraded broadcast channel is still an open problem even for the model without MAC. Our main result for this model is a single-letter characterization of the set of all achievable

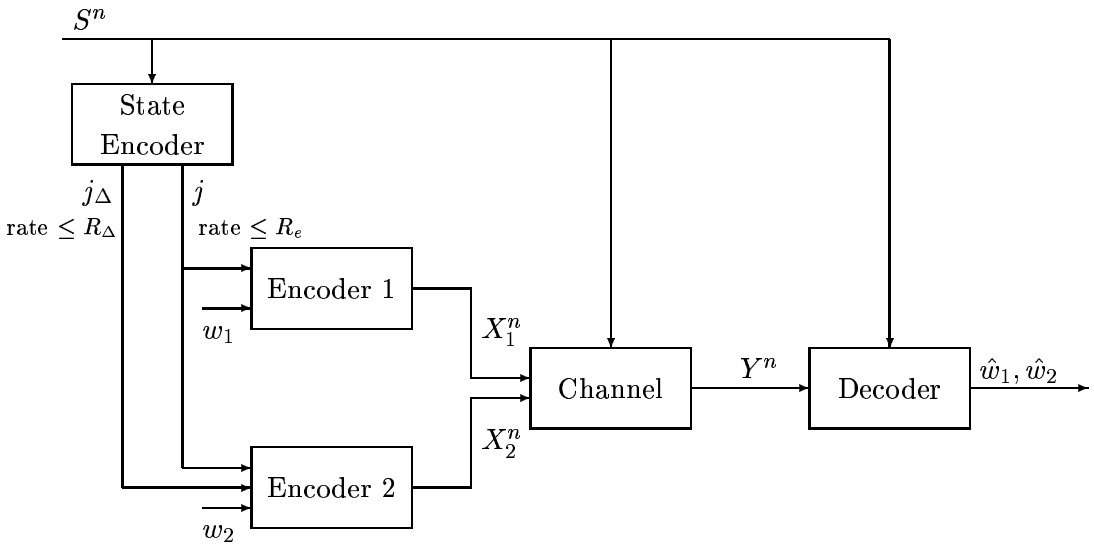


Figure 1: MAC with degraded partial CSIT and full CSIR

sextuples $(R_1, R_2, R_e, R_\Delta, \Gamma_1, \Gamma_2)$, where R_1 and R_2 stand for the forward transmission rates, and Γ_1 and Γ_2 stand for the input constraints of the two users.

The second model that we consider, depicted in Figure 2, is that of MAC with *distinct* partial CSIT and full CSIR. The difference between this model and the one described in Figure 1 is that, as its name suggests, we no longer impose the degraded structure of CSIT. That is, the state encoder produces two streams of codes CSI. The first stream, of rate R_{e1}, is intended to Encoder 1 only, and the second stream, of rate R_{e2}, is intended to Encoder 2 only. These streams need not be one a subset of the other, and in general, are distict. Accordingly, the CSIT provided to the encoders is not necessarily of a degraded structure. Our aim is again the region of achievable sextuples $(R_1, R_2, R_{e1}, R_{e2}, \Gamma_1, \Gamma_2)$. For this model we have not been able to fully characterize the region of achievable rates and costs. Instead, inner and outer bounds are provided.

This paper is organized as follows. The first part of the paper is devoted to the MAC with degraded partial CSIT and full CSIR. Notations and definitions are given in Section 2. The main result for the MAC with degraded partial CSIT and full CSIR is presented in Section 3. As an example, additive white gaussian noise (AWGN) MAC with fading is considered in Section 4. The proof of the main theorem is provided in Section 5. Finally, achievable region and an outer bound to the capacity region of the MAC with distinct partial CSIT and full CSIR is given in Section 6.

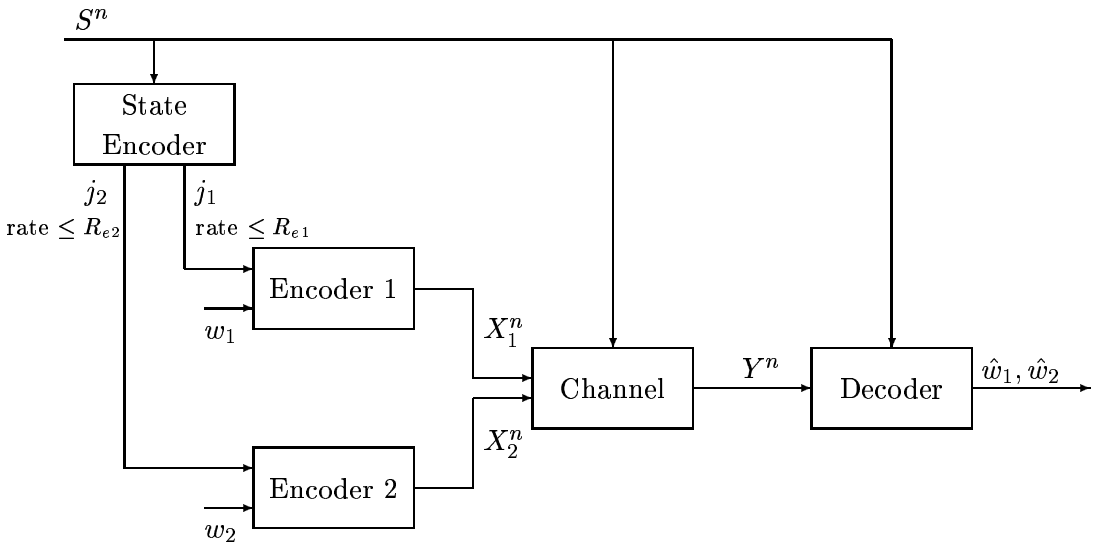


Figure 2: MAC with distinct partial CSIT and full CSIR

2 Notation and Preliminaries

Let $\mathcal{U}$ be a finite set, and let $\mathcal{U}^n$ stand for the set of all n-tuples with letters from $\mathcal{U}$. An element of $\mathcal{U}^n$ is written as $u^n = (u_1, u_2, \cdots, u_n)$. When $n = 1$, the superscript n will be omitted. The cardinality of a set $\mathcal{U}$ is denoted by $|\mathcal{U}|$. A convention that is kept throughout is that random variables are denoted by uppercase letters. Thus, random vectors in $\mathcal{U}^n$ are denoted by $U^n = (U_1, U_2, \cdots, U_n)$. Probability distributions are denoted by P and subscripted by the name of the random variables and the conditioning, e.g. $P_U(u)$ is the probability of $U = u$, $P_{UV}(u, v)$ is the joint probability of $U = u$, $V = v$, $P_{U|V}$ is the conditional probability of $U = u$ given $V = v$, etc. Sometimes, we will omit subscripts from the probability functions when it is clear from the context, e.g., $P(u, v)$ will sometimes denote $P_{UV}(u, v)$.

We follow the notation of [7] for information theoretic quantities such as entropy, mutual information, etc, and for typical sequences and typical sets. Thus, for a given pair of random variables (U, V) with joint distribution $P_{UV}(u, v)$, $H(U)$ denotes the entropy of U, $H(U, V)$ the joint entropy of (U, V), $H(U|V)$ the conditional entropy of U given V. The mutual

information is denoted by $I(U^n;V^n)$. Similarly, $A_\epsilon^n(U)$ is the set of ϵ-typical n-sequences

$$A_\epsilon^n(U) = \left\{ u^n \in \mathcal{U}^n : \left| \frac{1}{n} N(u|u^n) - P_U(u) \right| < \epsilon \right\}$$

where $N(u|u^n)$ is the number of occurrences of u in the vector u^n.

A discrete memoryless MAC controlled by memoryless, time-invariant state is a sextuple $(\mathcal{X}_1, \mathcal{X}_2, \mathcal{S}, P_S, P_{Y|X_1X_2S}, \mathcal{Y})$, where $\mathcal{X}_i$ is the input alphabet of user i, $i = 1, 2$, $\mathcal{S}$ is the state alphabet, P_S is the state probability mass function (PMF), $P_{Y|X_1X_2S}$ stands for the channel transition probability matrix, and $\mathcal{Y}$ is the output alphabet. The state and the channel are memoryless i.e.,

$$P_{S^n}(s^n) = \prod_{l=1}^{n} P_S(s_l) \tag{6}$$

$$P_{Y^n|X_1^nX_2^nS^n}(y^n|x_1^n, x_2^n, s^n) = \prod_{l=1}^{n} P_{Y|X_1X_2S}(y_l|x_{1l}, x_{2l}, s_l) \tag{7}$$

where $s^n = (s_1, \ldots, s_n) \in \mathcal{S}^n$, $x_1^n = (x_{11}, \ldots, x_{1n}) \in \mathcal{X}_1^n$ and similarly for $x_2^n \in \mathcal{X}_2^n$ and $y^n \in \mathcal{Y}^n$. When clear from the context, we will use the shorthand notation $P_{Y|X_1X_2S}$ for the channel.

The communication system that we explore here is depicted in Figure 1. A state sequence of length n, $s^n \in \mathcal{S}^n$, is generated and fed into the state encoder. The state encoder generates two rate-limited streams, the *common partial CSIT*, and the *incremental partial CSIT*. The common CSIT, whose index is denoted by j, is of rate R_e, and is intended to both encoders, Encoder 1 and Encoder 2. The incremental CSIT, whose index is denoted by j_Δ, is of rate R_Δ, and is intended to Encoder 2 only. These streams are sent to the intended receipients through noiseless way-side channels. Having the common partial CSIT j, the first encoder produces $x_1^n \in \mathcal{X}_1^n$ to encode the message w_1. The second user encodes w_2 to $x_2^n \in \mathcal{X}_2^n$ with the help of j and the incremental partial CSIT, j_Δ. Each transmitted input block x_i^n, $i = 1, 2$, is subject to additive normalized input constraint

$$\phi_i(x_i^n) \stackrel{\triangle}{=} \frac{1}{n} \sum_{l=1}^{n} \phi_i(x_{il}) \leq \Gamma_i \tag{8}$$

where $\phi_i : \mathcal{X}_i \to \mathbb{R}^+$ is a single-letter input cost function for user i, $i = 1, 2$. Both x_1^n and x_2^n are fed into the random state controlled memoryless MAC. Finally, the decoder, having access to the state sequence s^n, gives estimates of the messages w_1, w_2 from the channel

output $y^n \in \mathcal{Y}^n$. Since the CSIT available to encoder 1 is a subset of th CSIT available to encoder 2, we denote this scenario as *degraded partial CSIT*. The next definition formalizes this description.

Definition 1. An $(n, R_1, R_2, R_e, R_\Delta, \Gamma_1, \Gamma_2, \epsilon)$ degraded partial CSIT code for the random state controlled MAC $(\mathcal{X}_1, \mathcal{X}_2, \mathcal{S}, P_S, P_{Y|X_1X_2S}, \mathcal{Y})$, consists of first and second stage CSI mappings, f_s and $f_{s\Delta}$:

$$f_s \;:\; \mathcal{S}^n \to \{1, \ldots, 2^{nR_e}\} \tag{9}$$

$$f_{s\Delta} \;:\; \mathcal{S}^n \to \{1, \ldots, 2^{nR_\Delta}\} \tag{10}$$

user encoding functions f_1 and f_2:

$$f_1 \;:\; \{1, \ldots, 2^{nR_1}\} \times \{1, \ldots, 2^{nR_e}\} \to \mathcal{X}_1^n \tag{11}$$

$$f_2 \;:\; \{1, \ldots, 2^{nR_2}\} \times \{1, \ldots, 2^{nR_e}\} \times \{1, \ldots, 2^{nR_\Delta}\} \to \mathcal{X}_2^n \tag{12}$$

and a decoding function

$$g : \mathcal{Y}^n \times \mathcal{S}^n \to \{1, \ldots, 2^{nR_1}\} \times \{1, \ldots, 2^{nR_2}\} \tag{13}$$

such that

$$\phi_1(x_1^n(w_1, j)) \;=\; \frac{1}{n}\sum_{l=1}^{n} \phi_1(x_{1l}^n(w_1, j)) \le \Gamma_1 \tag{14}$$

$$\phi_2(x_2^n(w_2, j, j_\Delta)) \;=\; \frac{1}{n}\sum_{l=1}^{n} \phi_2(x_{2l}^n(w_2, j, j_\Delta)) \le \Gamma_2 \tag{15}$$

for $w_1 \in \{1, \ldots, 2^{nR_1}\}$, $w_2 \in \{1, \ldots, 2^{nR_2}\}$, $j \in \{1, \ldots, 2^{nR_e}\}$, $j_\Delta \in \{1, \ldots, 2^{nR_\Delta}\}$ and where the probability of error P_e satisfies

$$\begin{aligned} P_e \;&=\; 2^{-n(R_1+R_2)} \sum_{w_1=1}^{2^{nR_1}} \sum_{w_2=1}^{2^{nR_2}} \sum_{s^n} P_{S^n}(s^n) \cdot \\ &\qquad \sum_{y^n : g(y^n, s^n) \neq (w_1, w_2)} P_{Y^n|X_1^n X_2^n S^n}(y^n | f_1(w_1, f_s(s^n)), f_2(w_1, f_s(s^n), f_{s\Delta}(s^n)), s^n) \\ &\le\; \epsilon \end{aligned} \tag{16}$$

Given a pair $\Gamma = (\Gamma_1, \Gamma_2)$, the rate quadruple $(R_1, R_2, R_e, R_\Delta)$ is Γ-*achievable* if for any

$\epsilon > 0$ and sufficiently large n, there exits an $(n, R_1, R_2, R_e, R_\Delta, \Gamma_1, \Gamma_2, \epsilon)$ degraded partial CSIT code for $P_{Y|X_1X_2S}$. The closure of the set of all Γ-achievable rate quadruple is denoted by $\mathcal{R}(\Gamma)$. For a given side information rate pair (R_e, R_Δ) and input constraints $\Gamma = (\Gamma_1, \Gamma_2)$, the capacity region $\mathcal{C}(\Gamma, R_e, R_\Delta)$ is the closure of the set of all rate pairs (R_1, R_2) such that $(R_1, R_2, R_e, R_\Delta)$ is Γ-achievable.

In the following, we will examine also the capacity region where the two users receive exactly the same common partial CSIT (i.e. $R_\Delta = 0$). In the same way as in Definition 1, we can define an $(n, R_1, R_2, R_e, \Gamma_1, \Gamma_2, \epsilon)$ code for this case by setting $R_\Delta = 0$ in Definition 1. Moreover, we define $\mathcal{C}_c(\Gamma, R_e)$ as the capacity region with common CSIT for given R_e and $\Gamma = (\Gamma_1, \Gamma_2)$.

3 Main Results

The main result that we present in this section is a single letter characterization of $\mathcal{C}(\Gamma, R_e, R_\Delta)$. Given $\Gamma = (\Gamma_1, \Gamma_2)$ and (R_e, R_Δ), let $\mathcal{C}^*(\Gamma, R_e, R_\Delta)$ be the set of all rate pairs (R_1, R_2) for which there exist random variables $(S_0, S_\Delta, X_1, X_2)$ taking values in $\mathcal{S}_0 \times \mathcal{S}_\Delta \times \mathcal{X}_1 \times \mathcal{X}_2$ such that the following conditions hold simultaneously:

1. The following Markov chains hold

$$X_1 \leftrightarrow S_0 \leftrightarrow (S, S_\Delta, X_2) \tag{17}$$

$$X_2 \leftrightarrow (S_0, S_\Delta) \leftrightarrow (S, X_1) \tag{18}$$

$$(S_0, S_\Delta) \leftrightarrow (S, X_1, X_2) \leftrightarrow Y \tag{19}$$

2. The auxiliary alphabets $\mathcal{S}_0$ and $\mathcal{S}_\Delta$ satisfy

$$|\mathcal{S}_0| \leq |\mathcal{S}| + 6 \tag{20}$$

$$|\mathcal{S}_\Delta| \leq |\mathcal{S}|(|\mathcal{S}| + 6) + 5 \tag{21}$$

3. The partial description rates R_e and R_Δ are bounded by

$$R_e \geq I(S_0; S) \tag{22}$$

$$R_\Delta \geq I(S_\Delta; S|S_0) \tag{23}$$

4. The forward channel rates R_1 and R_2 are bounded by

$$R_1 \leq I(X_1; Y|X_2, S, S_0) \tag{24}$$

$$R_2 \leq I(X_2; Y|X_1, S, S_0, S_\Delta) \tag{25}$$

$$R_1 + R_2 \leq I(X_1, X_2; Y|S, S_0, S_\Delta) \tag{26}$$

5. Input constraint functions satisfy

$$E[\phi_i(X_i)] \leq \Gamma_i, \ i = 1, 2 \tag{27}$$

The main result of this section is as follows:

Theorem 1. *For any random state controlled discrete memoryless MAC with degraded partial CSIT,* $\mathcal{C}(\Gamma, R_e, R_\Delta) = \mathcal{C}^*(\Gamma, R_e, R_\Delta)$.

The proof is given in Section 5. We do not have to take the convex hull of $\mathcal{C}^*(\Gamma, R_e, R_\Delta)$ because $\mathcal{C}^*(\Gamma, R_e, R_\Delta)$ is indeed convex. The convexity of $\mathcal{C}^*(\Gamma, R_e, R_\Delta)$ is shown in Appendix A.

The single-user version of our problem was studied in [14], and in [22], [23]. There is a connection between coding of the CSI by the genie and classical rate-distortion theory. Indeed, the auxiliary random variable S_0 in (4) stands for the reproduction random variable for producing the source S at the channel encoder. However, instead of having a distortion measure, the quality of reproduction is measured by the resulting forward capacity, given by (2). Similar observations connecting the present results with the problem of hierarchical source coding ([16], [20], and [11]), can be made. In particular, (22) and (23) play the role of the rate equations of the coarse stage and refinement stage, respectively, in a successive

refinement problem. (In [16], [20], and [11], the cumulative rate $R_e + R_\Delta$ is used, instead of the incremental rate R_Δ, which we use in this work. For a comprehensive discussion on the difference between these two usages, see [26].) Again, there is no specific distortion measure for reproducing the source S at the two stages; instead, the quality of reproduction is measured by the resulting capacity region, expressed in (24), (25), and (26). This connection can be seen in the proof of the direct part of Theorem 1.

In order to determine the capacity region of the special case where the two users receive exactly the same common partial CSIT ($R_\Delta = 0$), dropping R_Δ from all the definitions given above, let $\mathcal{C}_c^*(\Gamma, R_e)$ be the set of all rate pairs (R_1, R_2) (given R_e and $\Gamma = (\Gamma_1, \Gamma_2)$) for which there exist random variables (S_0, X_1, X_2) taking values in $\mathcal{S}_0 \times \mathcal{X}_1 \times \mathcal{X}_2$ such that $\mathcal{C}_c^*(\Gamma, R_e)$ is obtained when S_Δ becomes degenerate in all the conditions of the definition of $\mathcal{C}^*(\Gamma, R_e, R_\Delta)$ and moreover condition 2 becomes $|\mathcal{S}_0| \leq |\mathcal{S}| + 5$ (eq. (21) is dropped).

Corollary 1. *For any random i.i.d. state controlled discrete memoryless MAC with common partial CSIT,* $\mathcal{C}_c(\Gamma, R_e) = \mathcal{C}_c^*(\Gamma, R_e)$.

Proof: The proof is based on Theorem 1. We start with the converse part. The random variable S_Δ appears only in the equation for the incremental rate, R_Δ, and in the forward rates bounds (25) and (26). Setting $R_\Delta = 0$ implies the Markov structure $S_\Delta \leftrightarrow S_0 \leftrightarrow S$. Now observe that due to (19), we have

$$I(X_2; Y|X_1, S, S_0, S_\Delta) \leq I(X_2; Y|X_1, S, S_0) \tag{28}$$

$$I(X_1, X_2; Y|S, S_0, S_\Delta) \leq I(X_1, X_2; Y|S, S_0) \tag{29}$$

which proves the converse. For the direct part, observe that we always can take S_Δ to be a degenerate random variable, independent of all other random variables in the system. This will result in the desired characterization. □

4 Application: AWGN MAC with Fading

Consider the discrete time channel

$$Y_n = S_{1n}X_{1n} + S_{2n}X_{2n} + W_n \tag{30}$$

where S_{1n} and S_{2n} are i.i.d. processes which represent fading coefficients of the users. $W_n \sim \mathcal{N}(0,1)$ is an i.i.d. process which represents AWGN independent of S_{1n}, S_{2n}, X_{1n} and X_{2n}. We assume also average power constraints $E[x_1^2] \leq \Gamma_1$ and $E[x_2^2] \leq \Gamma_2$ ($\phi_i = x_i^2$, $i = 1,2$). It is possible to extend $\mathcal{C}_c^*(\Gamma, R_e)$ to continuous alphabets [12, Chapter 7]. In the view of Corollary 1, taking $S = (S_1, S_2)$, we can write $\mathcal{C}_c(\Gamma, R_e)$ as the set of all rate pairs (R_1, R_2) such that

$$\begin{aligned} R_1 &\leq I(X_1; Y|X_2, S_1, S_2, S_0) \\ R_2 &\leq I(X_2; Y|X_1, S_1, S_2, S_0) \\ R_1 + R_2 &\leq I(X_1, X_2; Y|S_1, S_2, S_0) \\ R_e &\geq I(S_0; S_1, S_2) \\ E[X_i^2] &\leq \Gamma_i,\ i = 1,2 \end{aligned}$$

for some joint distribution

$$P_{X_1X_2YS_1S_2S_0}(x_1, x_2, y, s_1, s_2, s_0) =$$
$$P_{S_1S_2}(s_1s_2)P_{S_0|S_1S_2}(s_0|s_1, s_2)P_{X_1|S_0}(x_1|s_0)P_{X_2|S_0}(x_2|s_0)P_{Y|X_1X_2S_1S_2}(y|x_1, x_2, s_1, s_2)$$

We can expand $I(X_1; Y|X_2, S_1, S_2, S_0)$, $I(X_2; Y|X_1, S_1, S_2, S_0)$ and $I(X_1, X_2; Y|S_1, S_2, S_0)$ as

$$\begin{aligned} I(X_1; Y|X_2, S_1, S_2, S_0) &= \sum_{s_1,s_2,s_0} P_{S_1S_2S_0}(s_1, s_2, s_0)[h(Y|X_2, s_1, s_2, s_0) \\ &\qquad -h(Y|X_1, X_2, s_1, s_2, s_0)] \qquad (31) \\ I(X_2; Y|X_1, S_1, S_2, S_0) &= \sum_{s_1,s_2,s_0} P_{S_1S_2S_0}(s_1, s_2, s_0)[h(Y|X_1, s_1, s_2, s_0) \\ &\qquad -h(Y|X_1, X_2, s_1, s_2, s_0)] \qquad (32) \\ I(X_1, X_2; Y|S_1, S_2, S_0) &= \sum_{s_1,s_2,s_0} P_{S_1S_2S_0}(s_1, s_2, s_0)[h(Y|s_1, s_2, s_0) \\ &\qquad -h(Y|X_1, X_2, s_1, s_2, s_0)] \qquad (33) \end{aligned}$$

Then, for the common term in (31)-(33) consider

$$\begin{aligned} h(Y|X_1, X_2, s_1, s_2, s_0) &\stackrel{a}{=} h(s_1X_1 + s_2X_2 + W|X_1, X_2, s_1, s_2) \\ &= h(W|X_1, X_2, s_1, s_2) \stackrel{b}{=} h(W) = \frac{1}{2}\log(2\pi e) \qquad (34) \end{aligned}$$

where a follows from the fact that $S_0 \to (S_1, S_2, X_1, X_2) \to Y$ forms a Markov chain, and b since W and (X_1, X_2, S_1, S_2) are independent of each other. For every triple (s_1, s_2, s_0), consider also

$$h(Y|X_2, s_1, s_2, s_0) = h(s_1X_1 + s_2X_2 + W|X_2, s_1, s_2, s_0) = h(s_1X_1 + W|X_2, s_1, s_2, s_0)$$

At this point, we need to use

$$\begin{aligned} Var(s_1X_1 + W|X_2, s_1, s_2, s_0) &= s_1^2 Var(X_1|X_2, s_1, s_2, s_0) + Var(W) \\ &\stackrel{a}{=} s_1^2 Var(X_1|s_0) + 1 \leq s_1^2 E[X_1^2|s_0] + 1 = s^2\varepsilon_1(s_0) + 1 \end{aligned}$$

where $\varepsilon_1(s_0) \stackrel{\triangle}{=} E[X_1^2|s_0]$ and a follows from the fact that $(S, X_2) \to S_0 \to X_1$ forms a Markov chain. Due to the fact that for a given second moment, normal distribution maximizes differential entropy, we get

$$h(Y|X_2, s_1, s_2, s_0) \leq \frac{1}{2}\log(2\pi e(1 + s_1^2\varepsilon_1(s_0))) \tag{35}$$

Furthermore, applying similar arguments,

$$h(Y|X_1, s_1, s_2, s_0) \leq \frac{1}{2}\log(2\pi e(1 + s_2^2\varepsilon_2(s_0))) \tag{36}$$

$$h(Y|s_1, s_2, s_0) \leq \frac{1}{2}\log(2\pi e(1 + s_1^2\varepsilon_1(s_0) + s_2^2\varepsilon_2(s_0))) \tag{37}$$

where $\varepsilon_2(s_0) \stackrel{\triangle}{=} E[X_2^2|s_0]$. Substituting (34)-(37) into (31)-(33), for any fixed distribution $P_{S_0|S_1,S_2}(s_0|s_1, s_2)$, we get the following upper bounds

$$\begin{aligned} I(X_1; Y|X_2, S_1, S_2, S_0) &\leq \sum_{s_1,s_2} P_{S_1,S_2}(s_1, s_2) \sum_{s_0} P_{S_0|S_1,S_2}(s_0|s_1, s_2)\frac{1}{2}\log(1 + s_1^2\varepsilon_1(s_0)) \\ I(X_2; Y|X_1, S_1, S_2, S_0) &\leq \sum_{s_1,s_2} P_{S_1,S_2}(s_1, s_2) \sum_{s_0} P_{S_0|S_1,S_2}(s_0|s_1, s_2)\frac{1}{2}\log(1 + s_2^2\varepsilon_2(s_0)) \\ I(X_1, X_2; Y|S_1, S_2, S_0) &\leq \sum_{s_1,s_2} P_{S_1,S_2}(s_1, s_2) \cdot \\ &\qquad \sum_{s_0} P_{S_0|S_1,S_2}(s_0|s_1, s_2)\frac{1}{2}\log(1 + s_1^2\varepsilon_1(s_0) + s_2^2\varepsilon_2(s_0)) \end{aligned}$$

where the inequalities are satisfied with equality if $X_1|s_0 \sim \mathcal{N}(0, \varepsilon_1(s_0))$ and $X_2|s_0 \sim \mathcal{N}(0, \varepsilon_2(s_0))$ for every s_0. Hence, we have shown the following result

Theorem 2. *The capacity region $\mathcal{C}_c(\Gamma, R_e)$ for the AWGN MAC with fading given by (30) reduces to the set of all rate pairs that satisfy*

$$R_i \leq \sum_{s_i} P_{S_i}(s_i) \sum_{s_0} P_{S_0|S_i}(s_0|s_i) \frac{1}{2} \log(1 + s_i^2 \varepsilon_i(s_0)), \ i = 1, 2 \tag{38}$$

$$R_1 + R_2 \leq \sum_{s_1, s_2} P_{S_1,S_2}(s_1, s_2) \sum_{s_0} P_{S_0|S_1,S_2}(s_0|s_1, s_2) \frac{1}{2} \log(1 + s_1^2 \varepsilon_1(s_0) + s_2^2 \varepsilon_2(s_0)) \tag{39}$$

for some probability distribution $P_{S_0|S_1,S_2}(s_0|s_1, s_2)$ and power allocation functions $\varepsilon_1(s_0)$, $\varepsilon_2(s_0)$ such that

$$R_e \geq I(S_0; S_1, S_2) \tag{40}$$

$$E[\varepsilon_i(S_0)] \leq \Gamma_i, \ i = 1, 2 \tag{41}$$

and

$$|\mathcal{S}_0| \leq |\mathcal{S}_1||\mathcal{S}_2| + 5 \tag{42}$$

Finally, finding the capacity region of a general AWGN MAC with fading reduces to finding the optimum distribution $P_{S_0|S_1,S_2}(s_0|s_1, s_2)$ and the optimum power allocation functions $\varepsilon_1(s_0)$, $\varepsilon_2(s_0)$, satisfying the above constraints.

5 Proof of Theorem 1

5.1 Direct Part

We will prove that all rate pairs $(R_1, R_2) \in \mathcal{C}^*(\Gamma, R_e, R_\Delta)$ are achievable with input constraints $\Gamma = (\Gamma_1, \Gamma_2)$ and CSI coding rates (R_e, R_Δ). In our proof, the state encoding scheme is similar to that of hierarchical source coding (see [20]). In the encoding and decoding of messages, the random coding approach for usual MAC [7, Section 14.3] is adapted to our case. Consider the following coding scheme:

Codebook Generation

1. Randomly generate 2^{nR_e} vectors $s_0^n(j)$, drawn i.i.d. $\sim \prod_{l=1}^n P_{S_0}(s_{0l})$. Index these sequences by $j \in \{1, \ldots, 2^{nR_e}\}$.

2. For each codeword $s_0^n(j)$, generate 2^{nR_Δ} independent codewords $s_\Delta^n(j, j_\Delta)$, $j_\Delta \in \{1, \ldots, 2^{nR_\Delta}\}$,

according to $\sim \prod_{l=1}^{n} P_{S_\Delta|S_0}(s_{\Delta l}|s_{0l}(j))$.

3. For each codeword $s_0^n(j)$, generate 2^{nR_1} independent codewords $x_1^n(w_1, j)$, $w_1 \in \{1, \ldots, 2^{nR_1}\}$, according to $\sim \prod_{l=1}^{n} P_{X_1|S_0}(x_{1l}|s_{0l}(j))$.

4. For each code pair $(s_0^n(j), s_\Delta^n(j, j_\Delta))$, generate 2^{nR_2} independent codewords $x_2^n(w_2, j, j_\Delta)$, $w_2 \in \{1, \ldots, 2^{nR_2}\}$, according to $\sim \prod_{l=1}^{n} P_{X_2|S_0S_\Delta}(x_{1l}|s_{0l}(j), s_{\Delta l}(j, j_\Delta))$.

Reveal the codebooks to the state encoder, channel encoders, and decoder.

Encoding

1. *State Encoding*: When s^n is the state vector, it is encoded by (j, j_Δ) if there exists a pair (j, j_Δ) such that $(s^n, s_0^n(j), s_\Delta^n(j, j_\Delta)) \in A_\epsilon^n(S, S_0, S_\Delta)$. Denote them by $(j(s^n), j_\Delta(s^n))$. If there is no such (j, j_Δ), let $(j, j_\Delta) = (1, 1)$. If there is more than one pair, take the first pair according to the lexicographic order. Send j to the first encoder and (j, j_Δ) to the second encoder.

2. *Encoding (User 1)*: To send index w_1 when j is received from the state encoder, the first encoder sends $x_1^n(w_1, j)$.

3. *Encoding (User 2)*: To send index w_2 when (j, j_Δ) is received from the state encoder, the second encoder sends $x_2^n(w_2, j, j_\Delta)$.

Decoding

1. Having the state vector s^n, the decoder determines the partial CSI indices (j, j_Δ) that were sent to the channel encoders, applying the state encoding step. An error is declared if there is no pair (j, j_Δ) satisfying typicality as explained in the state encoding step.

2. If an error is not declared in the first part, the decoder chooses the unique pair $(\hat{w}_1, \hat{w}_2)$ such that

$$(s^n, s_0^n(j), s_\Delta^n(j, j_\Delta), y^n, x_1^n(\hat{w}_1, j), x_2^n(\hat{w}_2, j, j_\Delta)) \in A_\epsilon^n(S, S_0, S_\Delta, Y, X_1, X_2).$$

   If such a pair does not exist, or more than one such a pair exists, an error is declared. Also, the receiver declares an error if at least one of the chosen codewords exceeds the power constraints by γ_i, $i = 1, 2$, (according to (14)-(15)), that is if:

$$\phi_1(x_1^n(\hat{w}_1, j(s^n) \geq \Gamma_1 + \gamma_1(\epsilon)$$

or

$$\phi_2(x_2^n(\hat{w}_2, j(s^n), j_\Delta(s^n))) \geq \Gamma_2 + \gamma_2(\epsilon)$$

Analysis of Probability of Error

Define the following events:

$$\begin{aligned}
E_1(s^n) &= \{(s^n, s_0^n(j)) \notin A_\epsilon^n(S, S_0), \forall j \in \{1, \ldots, 2^{nR_e}\}\} \\
E_2(s^n) &= \{(s^n, s_0^n(j), s_\Delta^n(j, j_\Delta)) \notin A_\epsilon^n(S, S_0, S_\Delta), \\
&\qquad \forall (j, j_\Delta), j \in \{1, \ldots, 2^{nR_e}\}, j_\Delta \in \{1, \ldots, 2^{nR_\Delta}\}\} \\
E_3(w_1, w_2, s^n) &= \{(s^n, s_0^n(j), s_\Delta^n(j, j_\Delta), x_1^n(w_1, j), x_2^n(w_1, j, j_\Delta)) \notin A_\epsilon^n(S, S_0, S_\Delta, X_1, X_2)\} \\
E_4(w_1, w_2, s^n) &= \{(s^n, s_0^n(j), s_\Delta^n(j, j_\Delta), x_1^n(w_1, j), x_2^n(w_1, j, j_\Delta), y^n) \in A_\epsilon^n(S, S_0, S_\Delta, X_1, X_2, Y)\} \\
E_5(w_1, s^n) &= \{\phi_1(x_1^n(w_1, j(s^n))) > \Gamma_1 + \gamma_1(\epsilon)\} \\
E_6(w_2, s^n) &= \{\phi_2(x_2^n(w_2, j(s^n), j_\Delta(s^n))) > \Gamma_2 + \gamma_2(\epsilon)\}
\end{aligned}$$

Due to the symmetry of the random code construction, the conditional error probability does not depend on which message pair (w_1, w_2) is sent. Therefore, we can assume that $(w_1, w_2) = (1, 1)$ is sent. Then, the average error probability defined in (16) can be bounded by:

$$\begin{aligned}
P_e &= Pr((\hat{w}_1, \hat{w}_2) \neq (1,1) | (1,1) \text{ is sent}) \\
&\leq \sum_{s^n \notin A_\epsilon^n(S)} P_{S^n} \\
&\quad + \sum_{s^n \in A_\epsilon^n(S)} P_{S^n}(s^n)[P(E_1(s^n)) + P(E_2(s^n)|E_1^c(s^n)) \\
&\qquad + P(E_3(1,1,s^n)|E_1^c(s^n), E_2^c(s^n)) \\
&\qquad + P(E_4^c(1,1,s^n)|E_1^c(s^n), E_2^c(s^n), E_3^c(1,1,s^n)) \\
&\qquad + \sum_{w_1 \neq 1} P(E_4(w_1,1,s^n)|E_1^c(s^n), E_2^c(s^n), E_3^c(1,1,s^n)) \\
&\qquad + \sum_{w_2 \neq 1} P(E_4(1,w_2,s^n)|E_1^c(s^n), E_2^c(s^n), E_3^c(1,1,s^n)) \\
&\qquad + \sum_{w_1 \neq 1, w_2 \neq 1} P(E_4(w_1,w_2,s^n)|E_1^c(s^n), E_2^c(s^n), E_3^c(1,1,s^n)) \\
&\qquad + P(E_5(1,s^n)) + P(E_6(1,s^n))] \qquad (43)
\end{aligned}$$

1. When n is large enough, it is straightforward that

$$\sum_{s^n \notin A_\epsilon^n(S)} P_{S^n} < \epsilon \tag{44}$$

In the remaining steps of this analysis, we assume that $s^n \in A_\epsilon^n(S)$ according to (43).

2. From standard arguments, employed for example in rate distortion theory [7, Section 13.5],

$$P(E_1(s^n)|) < \epsilon \tag{45}$$

when $R_e > I(S;S_0) + \delta_1(\epsilon)$. Conditioning on $E_1^c(s^n)$ means that there exists j such that $(s^n, s_0^n(j)) \in A_\epsilon^n(S,S_0)$. Hence, again using arguments from rate distortion theory [10] (see also [20])

$$P(E_2(s^n)|E_1^c(s^n)) < \epsilon \tag{46}$$

when $R_\Delta \geq I(S;S_\Delta|S_0) + \delta_2(\epsilon)$ for sufficiently large n.

3. The conditioning on $E_1^c(s^n)$ and $E_2^c(s^n)$ means that there exist $s_0^n(j)$ and $s_\Delta^n(j,j_\Delta)$ such that $(s^n, s_0^n(j), s_\Delta^n(j,j_\Delta)) \in A_\epsilon^n(S,S_0,S_\Delta)$. Recall that in the construction of the codebook, the codewords $x_1^n(1,j)$ and $x_2^n(1,j,j_\Delta)$, are drawn according to $\prod_{l=1}^n P_{X_1|S_0}(x_{1l}|s_{0l}(j))$ and $\prod_{l=1}^n P_{X_2|S_0S_\Delta}(x_{1l}|s_{0l}(j), s_{\Delta l}(j,j_\Delta))$. Therefore

$$Pr((s_0^n(j), s_\Delta^n(j,j_\Delta), x_1^n(1,j), x_2^n(1,j,j_\Delta)) \in A_\epsilon^n(S_0,S_\Delta,X_1,X_2)) \to 1 \tag{47}$$

for sufficiently large n. Then, using the fact that $S \leftrightarrow (S_0,,S_\Delta) \leftrightarrow (X_1,X_2)$ forms Markov chain,

$$P(E_3(1,1,s^n)|E_1^c(s^n), E_2^c(s^n)) < \epsilon \tag{48}$$

for sufficiently large n using the Markov lemma [7, Lemma 14.8.1].

4. While conditioning on $E_3^c(1,1,s^n)$, we have $(s^n, s_0^n(j), s_\Delta^n(j,j_\Delta), x_1^n(1,j), x_2^n(1,j,j_\Delta)) \in A_\epsilon^n(S,S_0,S_\Delta,X_1,X_2)$. Furthermore, using the Markov structure $(S_0,S_\Delta) \leftrightarrow (S,X_1,X_2) \leftrightarrow$

Y and using again Markov Lemma

$$P(E_4^c(1,1,s^n)|E_1^c(s^n),E_2^c(s^n),E_3^c(1,1,s^n)) < \epsilon \tag{49}$$

for sufficiently large n.

5. Knowing that $s_0^n = s_0^n(j) = s_0^n(j(s^n))$ and $s_\Delta^n = s_\Delta^n(j,j_\Delta) = s_\Delta^n(j(s^n), j_\Delta(s^n))$, for any $w_1 \neq 1$ consider the following

$$\begin{aligned}
&P(E_4(w_1,1,s^n)|E_1^c(s^n),E_2^c(s^n),E_3^c(1,1,s^n)) \\
&= \sum_{(s^n,s_0^n,s_\Delta^n,y^n,x_1^n,x_2^n)\in A_\epsilon^n} [P_{S^nS_0^nS_\Delta^n}(s^n,s_0^n,s_\Delta^n)P_{X_1^n|S^nS_0^nS_\Delta^n}(x_1^n|s^n,s_0^n,s_\Delta^n)\cdot \\
&\qquad P_{Y^nX_2^n|S^nS_0^nS_\Delta^n}(y^n,x_2^n|s^n,s_0^n,s_\Delta^n)] \\
&\leq |A_\epsilon^n| 2^{-n(H(S,S_0,S_\Delta)-\epsilon)} 2^{-n(H(X_1|S,S_0,S_\Delta)-\epsilon)} 2^{-n(H(Y,X_2|S,S_0,S_\Delta)-\epsilon)} \\
&\leq 2^{n(H(X_1,X_2,Y,S,S_0,S_\Delta)+\epsilon)} 2^{-n(H(S,S_0,S_\Delta)-\epsilon)} 2^{-n(H(X_1|S,S_0,S_\Delta)-\epsilon)} 2^{-n(H(Y,X_2|S,S_0,S_\Delta)-\epsilon)} \\
&= 2^{-n(H(X_1|S,S_0,S_\Delta)-[H(X_1,X_2,Y,S,S_0,S_\Delta)-H(S,S_0,S_\Delta)-H(Y,X_2|S,S_0,S_\Delta)]-4\epsilon)} \\
&= 2^{-n(H(X_1|S,S_0,S_\Delta)-[H(X_1,X_2,Y|S,S_0,S_\Delta)-H(Y,X_2|S,S_0,S_\Delta]-4\epsilon)} \\
&= 2^{-n(H(X_1|S,S_0,S_\Delta)-H(X_1|Y,X_2,S,S_0,S_\Delta)-4\epsilon)} = 2^{-n(I(X_1;Y,X_2|S,S_0,S_\Delta)-4\epsilon)} \\
&\overset{(a)}{=} 2^{-n(I(X_1;Y|X_2,S,S_0,S_\Delta)-4\epsilon)} \\
&\overset{(b)}{=} 2^{-n(I(X_1;Y|X_2,S,S_0)-4\epsilon)}
\end{aligned} \tag{50}$$

where (a) follows from the fact that X_1 and X_2 are independent given (S_0, S_Δ) and $S \to (S_0, S_\Delta) \to (X_1, X_2)$, and (b) follows from the Markov relations (17) and (19). Hence, we have

$$\begin{aligned}
&\sum_{w_1\neq 1} P(E_4(w_1,1,s^n)|E_1^c(s^n),E_2^c(s^n),E_3^c(1,1,s^n)) \\
&\leq 2^{nR_1} 2^{-n(I(X_1;Y|X_2,S,S_0)-4\epsilon)} = 2^{-n(I(X_1;Y|X_2,S,S_0)-R_1-4\epsilon)} < \epsilon
\end{aligned} \tag{51}$$

for n sufficiently large when $R_1 < I(X_1;Y|X_2,S,S_0) - 4\epsilon$.

6. Applying similar steps, we have

$$\begin{aligned}
&\sum_{w_2\neq 1} P(E_4(1,w_2,s^n)|E_1^c(s^n),E_2^c(s^n),E_3^c(1,1,s^n)) \\
&\leq 2^{-n(I(X_2;Y|X_1,S,S_0,S_\Delta)-R_2-4\epsilon)} < \epsilon
\end{aligned} \tag{52}$$

for n sufficiently large when $R_2 < I(X_2; Y|X_1, S, S_0, S_\Delta) - 4\epsilon$.

7. For $w_1 \neq 1$, $w_2 \neq 1$ consider the following

$$
\begin{aligned}
&P(E_4(w_1, w_2, s^n)|E_1^c(s^n), E_2^c(s^n), E_4^c(1, 1, s^n)) \\
&= \sum_{(s^n, s_0^n, s_\Delta^n, y^n, x_1^n, x_2^n) \in A_\epsilon^n} [P_{S^n S_0^n S_\Delta^n}(s^n, s_0^n, s_\Delta^n) P_{X_1^n X_2^n | S^n S_0^n S_\Delta^n}(x_1^n, x_2^n | s^n, s_0^n, s_\Delta^n) \cdot \\
&\qquad P_{Y^n | S^n S_0^n S_\Delta^n}(y^n | s^n, s_0^n, s_\Delta^n)] \\
&\leq |A_\epsilon^n| 2^{-n(H(S,S_0,S_\Delta)-\epsilon)} 2^{-n(H(X_1,X_2|S,S_0,S_\Delta)-\epsilon)} 2^{-n(H(Y|S,S_0,S_\Delta)-\epsilon)} \\
&\leq 2^{n(H(X_1,X_2,Y,S,S_0,S_\Delta)+\epsilon)} 2^{-n(H(S,S_0,S_\Delta)-\epsilon)} 2^{-n(H(X_1,X_2|S,S_0,S_\Delta)-\epsilon)} 2^{-n(H(Y|S,S_0,S_\Delta)-\epsilon)} \\
&= 2^{-n(H(X_1,X_2|S,S_0,S_\Delta)-[H(X_1,X_2,Y,S,S_0,S_\Delta)-H(S,S_0,S_\Delta)-H(Y|S,S_0,S_\Delta)]-4\epsilon)} \\
&= 2^{-n(H(X_1,X_2|S,S_0,S_\Delta)-[H(X_1,X_2,Y|S,S_0,S_\Delta)-H(Y|S,S_0,S_\Delta]-4\epsilon)} \\
&= 2^{-n(H(X_1,X_2|S,S_0,S_\Delta)-H(X_1,X_2|Y,S,S_0,S_\Delta)-4\epsilon)} \\
&= 2^{-n(I(X_1,X_2;Y|S,S_0,S_\Delta)-4\epsilon)} \qquad (53)
\end{aligned}
$$

Then we have

$$
\begin{aligned}
&\sum_{w_1 \neq 1, w_2 \neq 1} P(E_4(w_1, w_2, s^n)|E_1^c(s^n), E_2^c(s^n), E_3^c(1, 1, s^n)) \qquad (54) \\
&\leq 2^{nR_1} 2^{nR_2} 2^{-n(I(X_1,X_2;Y|S,S_0,S_\Delta)-4\epsilon)} = 2^{-n(I(X_1,X_2;Y|S,S_0,S_\Delta)-(R_1+R_2)-4\epsilon)} < \epsilon \qquad (55)
\end{aligned}
$$

for n sufficiently large when $R_1 + R_2 < (X_1, X_2; Y|S, S_0, S_\Delta) - 4\epsilon$.

8. For the input constraint of the first user, we have by the weak law of large numbers,

$$
\begin{aligned}
P(E_5(1, s^n)) &= P(\phi_1(x_1^n(1, j(s^n))) > \Gamma_1 + \gamma_1(\epsilon)) \\
&= P\left(\frac{1}{n}\sum_{l=1}^{n} \phi_1(x_{1l}(1, j(s^n))) > \Gamma_1 + \gamma_1(\epsilon)\right) \\
&< \epsilon \qquad (56)
\end{aligned}
$$

for n large enough when $E[\phi_1(X_1)] < \Gamma_1$.

Following similar arguments,

$$P(E_6(1, s^n)) < \epsilon \qquad (57)$$

when $E[\phi_2(X_2)] < \Gamma_2$ for n large enough.

Finally, summing up the results of steps 1-8 in (43), we deduce that when

$$\begin{aligned} R_e &> I(S;S_0) + \delta_1(\epsilon) \\ R_\Delta &> I(S;S_\Delta|S_0) + \delta_2(\epsilon) \\ R_1 &< I(X_1;Y|X_2,S,S_0) - 4\epsilon \\ R_2 &< I(X_2;Y|X_1,S,S_0,S_\Delta) - 4\epsilon \\ R_1+R_2 &< I(X_1,X_2;Y|S,S_0,S_\Delta) - 4\epsilon \\ E[\phi_1(X_1)] &< \Gamma_1 \\ E[\phi_2(X_2)] &< \Gamma_2 \end{aligned}$$

then $\phi_i(X_i^n) \leq \Gamma_i + \gamma_i(\epsilon)$ with probability $1-\epsilon$ for $i=1,2$, and $P_e < 10\epsilon$ for n sufficiently large. In particular, note that

$$\begin{aligned} E[\phi_i(X_i^n)] &\leq (\Gamma_i + \gamma_i(\epsilon)) \cdot P(\phi_i(X_i^n) \leq \Gamma_i + \gamma_i(\epsilon)) \\ &\quad + P(\phi_i(X_i^n) > \Gamma_i + \gamma_i(\epsilon)) \cdot \max_{x_i} \phi_i(x_i) \\ &\leq \Gamma_i + \gamma_i(\epsilon) + \epsilon \cdot \max_{x_i} \phi_i(x_i) \end{aligned} \tag{58}$$

where in the last inequality, we have used (56) and (57). Since $\epsilon > 0$ can be made arbitrary small (with increasing block length) and $\gamma_i(\epsilon), \delta_i(\epsilon) \to 0$ when $\epsilon \to 0$, then $P_e \to 0$ when $(R_1,R_2) \in \mathcal{C}^*(\Gamma, R_e, R_\Delta)$ as $n \to \infty$. This completes the proof of the achievability part.

5.2 Converse Part

In this part, we will prove that, for a given pair $\Gamma = (\Gamma_1,\Gamma_2)$ and side information rate pair (R_e, R_Δ), any achievable sequence of codes $(n, R_1, R_2, R_e, R_\Delta, \Gamma_1, \Gamma_2, \epsilon)$ according to Definition 1 with $\epsilon \to 0$ should satisfy $(R_1,R_2) \in \mathcal{C}^*(\Gamma, R_e, R_\Delta)$. First, being (j, j_Δ) the output of the state encoder, let

$$S_{0l} = (j, S_l^+) \tag{59}$$

$$S_{\Delta l} = j_\Delta \tag{60}$$

where $S_l^+ = (S_{l+1}, S_{l+2}, \cdots, S_n)$. We note that S_{0l} and $S_{\Delta l}$ satisfy the Markov structures that are given in the condition 1 of the definition of $\mathcal{C}^*(\Gamma, R_e, R_\Delta)$:

$$X_{1l} \leftrightarrow S_{0l} \leftrightarrow (S_l, S_{\Delta l}, X_{2l}) \tag{61}$$

$$X_{2l} \leftrightarrow (S_{0l}, S_{\Delta l}) \leftrightarrow (S, X_{1l}) \tag{62}$$

$$(S_{0l}, S_{\Delta l}) \leftrightarrow (S_l, X_{1l}, X_{2l}) \leftrightarrow Y_l \tag{63}$$

Now, we have

$$\begin{aligned} nR_e &\geq H(j) \\ &= H(j) - H(j|S^n) = I(j; S^n) \\ &= \sum_{l=1}^{n} I(j; S_l|S_l^+) \\ &= \sum_{l=1}^{n} (I(j; S_l|S_l^+) + I(S_l; S_l^+)) \qquad (64) \\ &= \sum_{l=1}^{n} I(j, S_l^+; S_l) \\ &= \sum_{l=1}^{n} I(S_{0l}; S_l) \qquad (65) \end{aligned}$$

where

(64) follows since the state is memoryless

(65) from the definition of S_{0l} given by (59).

Also following similar steps, we have

$$\begin{aligned} nR_\Delta &\geq H(j_\Delta|j) \\ &= I(j_\Delta; S_n|j) \\ &= \sum_{l=1}^{n} I(j_\Delta; S_l|j, S_l^+) \\ &= \sum_{l=1}^{n} I(S_{\Delta l}; S_l|S_{0l}) \qquad (66) \end{aligned}$$

By the code construction, we estimate (w_1, w_2) using Y^n and S^n. Then, by Fano's inequality,

$$H(w_1|Y^n, S^n, j, j_\Delta) \leq H(w_1, w_2|Y^n, S^n, j, j_\Delta) \leq H(w_1, w_2|Y^n, S^n) \leq n\epsilon_n \quad (67)$$

$$H(w_2|Y^n, S^n, j, j_\Delta) \leq H(w_1, w_2|Y^n, S^n, j, j_\Delta) \leq H(w_1, w_2|Y^n, S^n) \leq n\epsilon_n \quad (68)$$

where $\epsilon_n \to 0$ when $P_e \to 0$. Therefore

$$\begin{aligned}
nR_1 &= H(w_1) && (69)\\
&= H(w_1|S^n, j_\Delta, j) && (70)\\
&= I(w_1; Y^n|S^n, j_\Delta, j) + H(w_1|Y^n, S^n, j_\Delta, j) && (71)\\
&\leq I(w_1; Y^n|S^n, j_\Delta, j) + n\epsilon_n && (72)
\end{aligned}$$

where (70) holds since the messages are independent of the state and (72) follows from (67). Then, consider

$$\begin{aligned}
nR_1 - n\epsilon_n &\leq I(X_1^n(w_1, j); Y^n|S^n, j_\Delta, j) && (73)\\
&= H(X_1^n(w_1, j)|S^n, j_\Delta, j) - H(X_1^n(w_1, j)|Y^n, S^n, j_\Delta, j) && (74)\\
&= H(X_1^n(w_1, j)|X_2^n(w_2, j_\Delta, j), S^n, j_\Delta, j) \\
&\quad - H(X_1^n(w_1, j)|Y^n, S^n, j_\Delta, j) && (75)\\
&\leq H(X_1^n(w_1, j)|X_2^n(w_2, j_\Delta, j), S^n, j_\Delta, j) \\
&\quad - H(X_1^n(w_1, j)|Y^n, X_2^n(w_2, j_\Delta, j), S^n, j_\Delta, j) && (76)\\
&= I(X_1^n(w_1, j); Y^n|X_2^n(w_2, j_\Delta, j), S^n, j_\Delta, j) && (77)\\
&= H(Y^n|X_2^n, S^n, j_\Delta, j) - H(Y^n|X_1^n, X_2^n, S^n, j_\Delta, j) && (78)\\
&= H(Y^n|X_2^n, S^n, j_\Delta, j) - H(Y^n|X_1^n, X_2^n, S^n) && (79)\\
&= \sum_{l=1}^{n} H(Y_l|Y_l^+, X_2^n, S^n, j_\Delta, j) - \sum_{l=1}^{n} H(Y_l|X_{1l}, X_{2l}, S_l) && (80)\\
&\leq \sum_{l=1}^{n} H(Y_l|X_{2l}, S_l, S_l^+, j, j_\Delta) - \sum_{l=1}^{n} H(Y_l|X_{1l}, X_{2l}, S_l) && (81)\\
&= \sum_{l=1}^{n} H(Y_l|X_{2l}, S_l, S_{0l}, S_{\Delta l}) - \sum_{l=1}^{n} H(Y_l|X_{1l}, X_{2l}, S_l) && (82)\\
&= \sum_{l=1}^{n} H(Y_l|X_{2l}, S_l, S_{0l}, S_{\Delta l}) - \sum_{l=1}^{n} H(Y_l|X_{1l}, X_{2l}, S_l, S_{0l}, S_{\Delta l}) && (83)
\end{aligned}$$

$$= \sum_{l=1}^{n} I(X_{1l}; Y_l | X_{2l}, S_l, S_{0l}, S_{\Delta l}) \quad (84)$$

$$= \sum_{l=1}^{n} I(X_{1l}; Y_l | X_{2l}, S_l, S_{0l}) \quad (85)$$

where

(73) follows from (72) and the Data Processing Lemma

(75) from the fact that X_1^n and X_2^n are independent given S^n, j and j_Δ.

(76) from adding conditioning

(79) from the fact that $(j, j_\Delta) \to (X_1^n, X_2^n, S^n) \to Y^n$ forms Markov Process

(80) from the memoryless property of the channel (see eq. (7)) and the chain rule

(81) from removing conditioning

(82) from the definitions given in (59) and (60)

(83) from the fact that $(S_0, S_\Delta) \to (S, X_1, X_2) \to Y$ forms a Markov chain

(85) from the Markov relations (61) and (63).

For the second rate, following the same steps, we obtain

$$nR_2 - n\epsilon_n \leq \sum_{l=1}^{n} I(X_{2l}; Y_l | X_{1l}, S_l, S_{0l}, S_{\Delta l}) \quad (86)$$

For the sum rate, using similar techniques, we have

$$n(R_1 + R_2) = H(w_1, w_2) \quad (87)$$

$$= H(w_1, w_2 | S^n, j_\Delta, j) \quad (88)$$

$$= I(w_1, w_2; Y^n | S^n, j_\Delta, j) + H(w_1, w_2 | Y^n, S^n, j_\Delta, j) \quad (89)$$

$$\leq I(w_1, w_2; Y^n | S^n, j_\Delta, j) + n\epsilon_n \quad (90)$$

where (90) is due to Fano's inequality. Continuing like the previous case:

$$
\begin{aligned}
n(R_1+R_2)-n\epsilon_n &\leq I(X_1^n(w_1,j),X_2^n(w_2,j_\Delta,j);Y^n|S^n,j_\Delta,j) && (91)\\
&= H(Y^n|S^n,j_\Delta,j)-H(Y^n|X_1^n,X_2^n,S^n,j_\Delta,j) && (92)\\
&= H(Y^n|S^n,j_\Delta,j)-H(Y^n|X_1^n,X_2^n,S^n) && (93)\\
&= \sum_{l=1}^{n} H(Y_l|Y_l^+,S^n,j_\Delta,j)-\sum_{l=1}^{n} H(Y_l|X_{1l},X_{2l},S_l) && (94)\\
&\leq \sum_{l=1}^{n} H(Y_l|S_l,S_l^+,j,j_\Delta)-\sum_{l=1}^{n} H(Y_l|X_{1l},X_{2l},S_l) && (95)\\
&= \sum_{l=1}^{n} H(Y_l|S_l,S_{0l},S_{\Delta l})-\sum_{l=1}^{n} H(Y_l|X_{1l},X_{2l},S_l) && (96)\\
&= \sum_{l=1}^{n} H(Y_l|S_l,S_{0l},S_{\Delta l})-\sum_{l=1}^{n} H(Y_l|X_{1l},X_{2l},S_l,S_{0l},S_{\Delta l}) && (97)\\
&= \sum_{l=1}^{n} I(X_{1l},X_{2l};Y_l|S_l,S_{0l},S_{\Delta l}) && (98)
\end{aligned}
$$

where

(91) follows from (90) and Data Processing Lemma

(93) from the fact that $(j_\Delta,j)\rightarrow(X_1^n,X_2^n,S^n)\rightarrow Y^n$ forms Markov Process

(94) from the memoryless property of the channel and the chain rule

(95) from removing conditioning

(96) from the definitions given in (59) and (60)

(97) from the fact that $(S_0,S_\Delta)\rightarrow(S,X_1,X_2)\rightarrow Y$ forms Markov Process

We turn now to the input constraints. By Definition 1, the code satisfies

$$
\begin{aligned}
\phi_1(x_1^n(w_1,j)) &= \frac{1}{n}\sum_{l=1}^{n}\phi_1(x_{1l}(w_1,j))\leq\Gamma_1 && (99)\\
\phi_2(x_2^n(w_2,j_\Delta,j)) &= \frac{1}{n}\sum_{l=1}^{n}\phi_2(x_{2l}(w_2,j_\Delta,j))\leq\Gamma_2 && (100)
\end{aligned}
$$

for $w_1\in\{1,\ldots,2^{nR_1}\}$, $j\in\{1,\ldots,2^{nR_e}\}$ and $j_\Delta\in\{1,\ldots,2^{nR_\Delta}\}$. We begin to look at the input constraint of the first user. Since each codeword satisfies the input constraint, their

average over w_1 and j also satisfies the input constraint. Thus, we can write

$$\begin{aligned}
\Gamma_1 &\geq \sum_{j=1}^{2^{nR_e}} \sum_{w_1=1}^{2^{nR_1}} \sum_{s^n} P_{S^n}(s^n) P(j|s^n) P(w_1) \phi_1(x_1^n(w_1, j)) \\
&= \sum_{j=1}^{2^{nR_e}} \sum_{w_1=1}^{2^{nR_1}} \sum_{s^n} P(s^n) P(j|s^n) P(w_1) \frac{1}{n} \sum_{l=1}^{n} \phi_1(x_{1l}(w_1, j)) \\
&= \sum_{x^n} \sum_{j=1}^{2^{nR_e}} \sum_{w_1=1}^{2^{nR_1}} \sum_{s^n} P(x^n|j, s^n, w_1) P_{S^n}(s^n) P(j|s^n) P(w_1) \frac{1}{n} \sum_{l=1}^{n} \phi_1(x_{1l}(w_1, j)) \\
&= \frac{1}{n} \sum_{l=1}^{n} E_{X_{1l}} \phi_1(X_{1l}) \qquad (101)
\end{aligned}$$

We turn now to the input constraint of the second user. Again, since each codeword of the second user satisfies input constraint according to the definition if the code, averaging over w_2, j and j_Δ, also satisfies the input constraint. Applying similar steps, we obtain the following:

$$\Gamma_2 \geq \frac{1}{n} \sum_{l=1}^{n} E_{X_{2l}} \phi_1(X_{2l}) \qquad (102)$$

Let Q be time sharing random variable which is uniformly distributed over $\{1, \ldots, n\}$. Substituting Q to (65), (66), (85), (86), (98), (101) and (102) we get

$$R_e \geq \frac{1}{n} \sum_{q=1}^{n} I(S_{0q}; S_q|Q = q) = I(S_{0Q}; S_Q|Q) = I(S_0; S|Q) \qquad (103)$$

$$R_\Delta \geq \frac{1}{n} \sum_{q=1}^{n} I(S_{\Delta q}; S_q|S_{0q}, Q = q) = I(S_{\Delta Q}; S_Q|S_{0Q}, Q) = I(S_\Delta; S|S_0, Q) \qquad (104)$$

and similarly

$$R_1 - \epsilon_n \leq I(X_1; Y|X_2, S, S_0, Q) \qquad (105)$$

$$R_2 - \epsilon_n \leq I(X_2; Y|X_1, S, S_0, S_\Delta, Q) \qquad (106)$$

$$R_1 + R_2\epsilon_n \leq I(X_1, X_2; Y|S, S_0, S_\Delta, Q) \qquad (107)$$

$$\Gamma_i \geq \sum_{q=1}^{n} \frac{1}{n} \sum_{x_i} P_{X_i|Q}(x_i|q) \phi_i(x_i) = E[\phi_i(X_i)], \ i = 1, 2 \qquad (108)$$

where $A \stackrel{\triangle}{=} A_Q$ for any random variable A that appears in (103)-(108). Finally, using the fact that $\mathcal{C}^*(\Gamma, R_e, R_\Delta)$ is convex (convexity of $\mathcal{C}^*(\Gamma, R_e, R_\Delta)$ is proved in Appendix A), (103)-

(108) imply that $(R_1, R_2) \in \mathcal{C}^*(\Gamma, R_e, R_\Delta)$ when $n \to \infty$.

To finish the converse part, we need to show that the alphabet sizes of auxiliary variables, $|\mathcal{S}_0|$ and $|\mathcal{S}_\Delta|$, can be upper-bounded as in the condition 2 (eq. (20) and (21)) of the definition of $\mathcal{C}^*(\Gamma, R_e, R_\Delta)$. This is done in Appendix B and this completes the proof.

6 Distinct Partial CSIT

In a more general setting, one can think of a setup where the two users receive distinct, non-degraded, but correlated partial CSIT. The communication system that we will examine in this section is depicted in Figure 2. The channel and the state model are the same as in Section 2. However, the state encoder produces two distinct rate-limited partial CSI streams, j_1 and j_2, with rates R_{e1} and R_{e2} respectively. Having appropriate *distinct partial CSIT* at each user, the encoding and the decoding of the messages are similar to the degraded partial CSIT case. In this setting, CSIT of one of the users is not necessarily a subset of the CSIT of the another one. In that sense, it is a more general setting than the previous one. For simplicity, we assume in this section that there are no input constraints. However, it is possible to add the constraints following similar techniques of the degraded CSIT case.

Definition 2. An $(n, R_1, R_2, R_{e1}, R_{e2}, \epsilon)$ code for the random state controlled MAC defined by the sextuple $(\mathcal{X}_1, \mathcal{X}_2, \mathcal{S}, P_S, P_{Y|X_1X_2S}, \mathcal{Y})$, using distinct partial CSIT consists of two mappings f_{s1} and f_{s2} of CSI:

$$f_{si} : \mathcal{S}^n \to \{1, \ldots, 2^{nR_{ei}}\},\ i = 1, 2 \tag{109}$$

user encoding functions f_1 and f_2:

$$f_i : \{1, \ldots, 2^{nR_i}\} \times \{1, \ldots, 2^{nR_{ei}}\} \to \mathcal{X}_i^n,\ i = 1, 2 \tag{110}$$

and decoding function

$$g : \mathcal{Y}^n \times \mathcal{S}^n \to \{1, \ldots, 2^{nR_1}\} \times \{1, \ldots, 2^{nR_2}\} \tag{111}$$

such that average error probability satisfies

$$\begin{aligned} P_e &= 2^{-n(R_1+R_2)} \sum_{w_1=1}^{2^{nR_1}} \sum_{w_2=1}^{2^{nR_2}} \sum_{s^n} P_{S^n}(s^n) \cdot \\ &\quad \sum_{y^n : g(y^n, s^n) \neq (w_1, w_2)} P_{Y^n|X_1^n X_2^n S^n}(y^n | f_1(w_1, f_{s1}(s^n)), f_2(w_2, f_{s2}(s^n)), s^n) \qquad (112) \\ &\leq \epsilon \end{aligned}$$

The rate quadruple $(R_1, R_2, R_{e1}, R_{e2})$ is said to be achievable if for any $\epsilon > 0$ and sufficiently large n, there exits an $(n, R_1, R_2, R_{e1}, R_{e2}, \epsilon)$ code. The rate region is the closure of the set of all achievable rate quadruples $(R_1, R_2, R_{e1}, R_{e2})$ and is denoted by $\mathcal{R}_d$.

We could not find complete single letter characterization of the achievable region. However, we give an achievable region and an outer bound. Define $\mathcal{R}_i$ as the convex hull of the set of all rate quadruples $(R_1, R_2, R_{e1}, R_{e2})$ for which there exist random variables (S_1, S_2, X_1, X_2) taking values in $\mathcal{S}_1 \times \mathcal{S}_2 \times \mathcal{X}_1 \times \mathcal{X}_2$ such that the following conditions hold simultaneously:

1. The following Markov chains hold

$$X_1 \leftrightarrow S_1 \leftrightarrow (S, S_2, X_2) \qquad (113)$$

$$X_2 \leftrightarrow S_2 \leftrightarrow (S, S_1, X_1) \qquad (114)$$

$$(S_1, S_2) \leftrightarrow (S, X_1, X_2) \leftrightarrow Y \qquad (115)$$

2. The partial description rates R_{e1} and R_{e2} satisfy

$$R_{ei} \geq I(S_i; S), \ i = 1, 2 \qquad (116)$$

$$R_{e1} + R_{e2} \geq I(S_1, S_2; S) + I(S_1; S_2) \qquad (117)$$

3. The forward channel rates R_1 and R_2 satisfy

$$R_1 \leq I(X_1; Y | X_2, S, S_1, S_2) \qquad (118)$$

$$R_2 \leq I(X_2; Y | X_1, S, S_1, S_2) \qquad (119)$$

$$R_1 + R_2 \leq I(X_1, X_2; Y | S, S_1, S_2) \qquad (120)$$

Define also $\mathcal{R}_o$ as the set of all rate quadruples $(R_1, R_2, R_{e1}, R_{e2})$ for which there exist random variables (S_1, S_2, X_1, X_2) taking values in $\mathcal{S}_1 \times \mathcal{S}_2 \times \mathcal{X}_1 \times \mathcal{X}_2$ such that all the conditions in the definition of $\mathcal{R}_i$ hold, except that (117) is replaced with

$$R_{e1} + R_{e2} \geq I(S_1, S_2; S) \tag{121}$$

We do not have to take convex hull of the set $\mathcal{R}_o$ because it is convex. The proof of the convexity is similar with the proof of the convexity of $\mathcal{C}^*(\Gamma, R_e, R_\Delta)$ and it is therefore omitted. We have the following theorem for the distinct CSIT case:

Theorem 3. *For any i.i.d. state controlled discrete memoryless MAC with distinct partial CSIT,* $\mathcal{R}_i \subseteq \mathcal{R}_d \subseteq \mathcal{R}_o$.

The proof of Theorem 3 is omitted here since it closely follows the proof of Theorem 1. The gap between the direct and converse part of Theorem 3 appears only in the bound of the sum of the partial description rates $R_{e1} + R_{e2}$. The coding scheme for partial distinct CSIT, achieving the inner bound in Theorem 3, has similarities with the coding scheme for the multiple description source coding problem without common encoder (see [10] for details). The complete characterization of achievable rates for the multiple description problem is still unsolved.

A Appendix A

Convexity of $\mathcal{C}^*(\Gamma, R_e, R_\Delta)$

In this section, we show that $\mathcal{C}^*(\Gamma, R_e, R_\Delta)$ is convex. For this purpose, let the set A be the set of all sextuples $(R_1, R_2, R_e, R_\Delta, \Gamma_1, \Gamma_2)$ such that

1. $R_e \geq I(S_0; S)$
   $R_\Delta \geq I(S_\Delta; S|S_0)$
2. $R_1 \leq I(X_1; Y|X_2, S, S_0)$
   $R_2 \leq I(X_2; Y|X_1, S, S_0, S_\Delta)$
   $R_1 + R_2 \leq I(X_1, X_2; Y|S, S_0, S_\Delta)$
3. $E[\phi_1[X_i]] \leq \Gamma_i$, $i = 1, 2$

where

$$P_{X_1X_2YSS_0S_\Delta Q}(x_1,x_2,y,s,s_0,s_\Delta) =$$
$$P_S(s)P_{S_0S_\Delta|S}(s_0,s_\Delta|s)P_{X_1|S_0}(x_1|s_0)P_{X_2|S_0S_\Delta}(x_2|s_0,s_\Delta)P_{Y|X_1X_2S}(y|x_1,x_2,s)$$

Define also B as the set of all sextuples $(R_1,R_2,R_e,R_\Delta,\Gamma_1,\Gamma_2)$ such that

$$\begin{aligned}
R_e &\geq I(S_0;S|Q) = I(S_0,Q;S) - I(S;Q) = I(S_0^*;S) && \text{(A-1)}\\
R_\Delta &\geq I(S_\Delta;S|S_0,Q) = I(S_\Delta;S|S_0^*) && \text{(A-2)}\\
R_1 &\leq I(X_1;Y|X_2,S,S_0,Q) = I(X_1;Y|X_2,S,S_0^*) && \text{(A-3)}\\
R_2 &\leq I(X_2;Y|X_1,S,S_0,S_\Delta,Q) = I(X_2;Y|X_1,S,S_0^*,S_\Delta) && \text{(A-4)}\\
R_1+R_2 &\leq I(X_1,X_2;Y|S,S_0,S_\Delta,Q) = I(X_1,X_2;Y|S,S_0^*,S_\Delta) && \text{(A-5)}\\
E[\phi_1[X_i]] &\leq \Gamma_i,\ i=1,2 && \text{(A-6)}
\end{aligned}$$

where

$$P_{X_1X_2YSS_0S_\Delta Q}(x_1,x_2,y,s,s_0,s_\Delta,q) =$$
$$P_S(s)P_Q(q)P_{S_0S_\Delta|SQ}(s_0,s_\Delta|s,q)P_{X_1|S_0Q}(x_1|s_0,q)P_{X_2|S_0S_\Delta Q}(x_2|s_0,s_\Delta,q)P_{Y|X_1X_2S}(y|x_1,x_2,s)$$

As a result, (A-1)-(A-6) imply that $(R_1,R_2,R_e,R_\Delta,\Gamma_1,\Gamma_2)\in A$ by introducing $S_0^* \stackrel{\triangle}{=} (S_0,Q)$. Note that S_0^* satisfies all the Markov conditions stated in the definition of $\mathcal{C}^*(\Gamma,R_e,R_\Delta)$ when we use it instead of S_0. Hence, we showed that $B\subseteq A$. Furthermore due to the fact that $B=CH(A)$ ($CH(\cdot)$ stands for the convex hull), $B=A$; proving the convexity of A.

Using the fact that, for a given pair $\Gamma=(\Gamma_1,\Gamma_2)$ and pair (R_e,R_Δ), $\mathcal{C}^*(\Gamma,R_e,R_\Delta)$ is a projection of A; we deduce that $\mathcal{C}^*(\Gamma,R_e,R_\Delta)$ is also a convex set.

B Appendix B

Alphabet Sizes of $\mathcal{S}_0$ and $\mathcal{S}_\Delta$

To bound the alphabet sizes of the auxiliary random variables in the definition of $\mathcal{C}^*(\Gamma,R_e,R_\Delta)$, $|\mathcal{S}_0|$ and $|\mathcal{S}_\Delta|$, we first recall the support lemma (cf. [8, p. 310] and [1, Lemma 3]). Let $\mathcal{P}(\mathcal{X})$ be the set of all probability distributions on an alphabet $\mathcal{X}$. Given continuous functionals f_m, $m=1,\ldots,k$, on $\mathcal{P}(\mathcal{X})$ and given any probability measure μ on the Borel σ-algebra of

$\mathcal{P}(\mathcal{X})$, there exists k elements, $Q_1, \ldots, Q_k$ of $\mathcal{P}(\mathcal{X})$ and constants $\alpha_l \geq 0$, $l = 1, \ldots, k$ with $\sum_{l=1}^{k} \alpha_l = 1$ such that

$$\int_{\mathcal{P}(\mathcal{X})} f_m(Q)\mu(dQ) = \sum_{l=1}^{k} \alpha_l f_m(Q_l),\ m = 1, \ldots, k \tag{B-1}$$

We first reduce the alphabet size of $\mathcal{S}_0$ while assuming that the alphabet size of $\mathcal{S}_\Delta$ is constant at this step. Being $P_{Y|X_1X_2S}$ the channel distribution and P_S the state source distribution, define the following functionals of a generic distribution Q over $\mathcal{X}_1 \times \mathcal{X}_2 \times \mathcal{Y} \times \mathcal{S} \times \mathcal{S}_\Delta$ where we assume that $\mathcal{S} = \{1, \ldots, p\}$, $p \stackrel{\triangle}{=} |\mathcal{S}|$:

$$f_s(Q) = \sum_{x_1, x_2, y, s_\Delta} Q(x_1, x_2, y, s, s_\Delta),\ s \in \{1, \ldots, p-1\} \tag{B-2}$$

$$f_p(Q) = H(S) - A(Q) \tag{B-3}$$

$$f_{p+1}(Q) = A(Q) - \sum_{x_1, x_2, y, s, s_\Delta} Q(x_1, x_2, y, s, s_\Delta) \log \frac{\sum_{x_1', x_2', y'} Q(x_1', x_2', y', s, s_\Delta)}{\sum_{x_1', x_2', y', s'} Q(x_1', x_2', y', s', s_\Delta)} \tag{B-4}$$

where

$$A(Q) = -\sum_{x_1, x_2, y, s, s_\Delta} Q(x_1, x_2, y, s, s_\Delta) \log \left(\sum_{x_1', x_2', y', s_\Delta'} Q(x_1', x_2', y', s, s_\Delta') \right) \tag{B-5}$$

Next, define

$$f_{p+2}(Q) = \sum_{x_1, x_2, y, s, s_\Delta} Q(x_1, x_2, y, s, s_\Delta) \log \frac{\sum_{x_1'} Q(x_1', x_2, y, s, s_\Delta)}{\sum_{x_1', y'} Q(x_1', x_2, y', s, s_\Delta)} + B(Q) \tag{B-6}$$

$$f_{p+3}(Q) = \sum_{x_1, x_2, y, s, s_\Delta} Q(x_1, x_2, y, s, s_\Delta) \log \frac{\sum_{x_2'} Q(x_1, x_2', y, s, s_\Delta)}{\sum_{x_2', y'} Q(x_1, x_2', y', s, s_\Delta)} + B(Q) \tag{B-7}$$

$$f_{p+4}(Q) = \sum_{x_1, x_2, y, s, s_\Delta} Q(x_1, x_2, y, s, s_\Delta) \log \frac{\sum_{x_1', x_2'} Q(x_1', x_2', y, s, s_\Delta)}{\sum_{x_1', x_2', y'} Q(x_1', x_2', y', s, s_\Delta)} + B(Q) \tag{B-8}$$

where

$$B(Q) = \sum_{x_1, x_2, y, s, s_\Delta} Q(x_1, x_2, y, s, s_\Delta) \log \frac{Q(x_1, x_2, y, s, s_\Delta)}{\sum_{y'} Q(x_1, x_2, y', s, s_\Delta)} \tag{B-9}$$

Finally, we define

$$f_{p+5}(Q) = \sum_{x_1,x_2,y,s_\Delta} Q(x_1,x_2,y,s,s_\Delta)\phi_1(x_1) \quad \text{(B-10)}$$

$$f_{p+6}(Q) = \sum_{x_1,x_2,y,s_\Delta} Q(x_1,x_2,y,s,s_\Delta)\phi_2(x_2) \quad \text{(B-11)}$$

Applying now the support lemma, we deduce that there exists a random variable S_0 (jointly distributed with X_1, X_2, Y, S and S_Δ satisfying Markov conditions in (17)-(19)) whose alphabet size is $|\mathcal{S}_0| = p+6 = |\mathcal{S}|+6$ that satisfies simultaneously

$$\begin{aligned}
\sum_{s_0} P(S_0=s_0) f_s(P(\cdot|s_0)) &= P(s),\ s=\{1,\ldots,p-1\} \\
\sum_{s_0} P(S_0=s_0) f_p(P(\cdot|s_0)) &= I(S_0;S) \\
\sum_{s_0} P(S_0=s_0) f_{p+1}(P(\cdot|s_0)) &= I(S_\Delta;S|S_0) \\
\sum_{s_0} P(S_0=s_0) f_{p+2}(P(\cdot|s_0)) &= I(X_1;Y|X_2,S,S_0) \\
\sum_{s_0} P(S_0=s_0) f_{p+3}(P(\cdot|s_0)) &= I(X_2;Y|X_1,S,S_0,S_\Delta) \\
\sum_{s_0} P(S_0=s_0) f_{p+4}(P(\cdot|s_0)) &= I(X_1,X_2;Y|S,S_0,S_\Delta) \\
\sum_{s_0} P(S_0=s_0) f_{p+5}(P(\cdot|s_0)) &= E[\phi_1(X_1)] \\
\sum_{s_0} P(S_0=s_0) f_{p+6}(P(\cdot|s_0)) &= E[\phi_2(X_2)]
\end{aligned}$$

Next, we proceed to upper bound the alphabet size of S_Δ. In a similar manner, we obtain $|\mathcal{S}||\mathcal{S}_0|-1$ constraints to fix the joint distribution of (S,S_0) and six other constraints to preserve $I(S_\Delta;S|S_0)$, $I(X_1;Y|X_2,S,S_0)$, $I(X_2;Y|X_1,S,S_0,S_\Delta)$, $I(X_1,X_2;Y|S,S_0,S_\Delta)$, $E[\phi_1(X_1)]$ and $E[\phi_2(X_2)]$. It is import ant to note that $I(S_0;S)$ has already preserved by fixing the joint distribution of (S,S_0). Thus using the support lemma, the alphabet size of S_Δ can be upper bounded by

$$|\mathcal{S}_\Delta| \leq |\mathcal{S}||\mathcal{S}_0|+5 \leq |\mathcal{S}|(|\mathcal{S}|+6)+5. \quad \text{(B-12)}$$

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